Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/13/2014
PI Name:	Wu, Honglu Ph.D.		
Project Title:	Dependence of radiation quality on charged particle-induced early and late damages in chromosomes		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per S. Monk/LaRC (Ed., 9/9/16) NOTE: End date changed to 12/31/2016 per L. Simonsen/LaRC (Ed., 12/31/15) NOTE: End date changed to 12/31/2015 per PI (Ed., 11/13/14)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Zhang, Ye (NASA Johnson Space Center)		
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Performance Goal Text:			

Task Description:	Tumor formation in humans or animals is a multi-step process. An early stage of cancer development is believed to be genomic instability (GI) which accelerates the mutation rate in the descendants of the cells surviving radiation exposure. A relationship between GI and tumor metastasis, as well as between GI and adaptation to drug treatment, has also been suggested. Evidence for GI induced from radiation exposure has been established in the last two decades. GI is defined as elevated or persistent genetic damages occurring many generations after the cells are exposed, and can be manifested in several biological consequences including mutation, apoptosis and damages in chromosomes. Genomic instability can be induced by both low- and high-LET radiation. Space radiation consists of a mixture of different types of ions covering a wide range of masses and energies. Its associated health risks to astronauts in flight are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. Before such missions can be undertaken, a much more detailed understanding of these biological effects is required to allow planning and implementation of protective countermeasures. The proposed research will attempt to investigate GI induced from exposures to energetic ions of the types and energies encountered during spaceflight, and will address the critical question of carcinogenesis risk and its dependence on radiation quality, which is an emphasized area of the current NASA Research Announcement (NRA). The overall goals of this proposed project are (1) to determine the relationship between the quality and fluence of energetic charged particles and the induction of delayed damages in chromosomes, which is commonly referred to as chromosomal instability and (2) to understand the relationship between chromosomal instability and other events at the cellular and molecular level that play a role in the maintenance of genome integrity. For this proposal, we select well-developed and widely used methodologies for quantification of GI. The three models include human T-lymphocytes collected from healthy subjects, and haemopoietic cells extracted from CBA/CaH and from C57BL/6J mice, which differ in radiosensitivity. The cells will be exposed in vitro to protons, Si, and Fe ions which represent three of the most important particle types in space. Chromosomal instability will be analyzed in both the total population of the cells surviving after multiple generations post irradiation (non-clonal assay), as well as in clonal populations derived from single cells surviving particle irradiation (clonal assay). The non-clonal assay is less labor intensive and allows analysis of samples exposed to particles of different types, energies, and doses, whereas the clonal assay will be designed to tackle the question of genetic background influence on the induction of GI. Several techniques for chromosome analysis will be used which include multi-color fluorescence in situ hybridization (mFISH) and the multi-color banding in situ hybridization (mBAND). Measurements of telomere length and expressions of genes involved in genomic stability will also be performed in cells from unstable, as well as stable, clones. The clonal expansion assay is arguably more suitable than the non-clonal assay for determining the correlation between chromosomal instability and other biological consequences. Key to our proposal is that the investigators have substantial expertise and published literature to ensure the success of the proposed studies. The PI has extensive experience in the induction of heavy ion-induced chromosomal aberrations in human lymphocytes using mFISH, mBAND, and other modern chromosome painting techniques. The Co-I was trained in immunology and has substantial expertise in T-lymphocyte isolation and expansion. Dr. Kadhim is a recognized expert in GI and had made significant contributions to the understanding of radiation-induced GI.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research addresses the question of genomic instability from charged particle exposures in space. By assessing genomic instability quantitatively for different charges and energies of the particles, we aim at contributing to the reduction of uncertainties in the assessment of cancer risks in astronauts after long-duration spaceflights.
Task Progress:	For investigations of genomic instability, bone marrow cells obtained from CBA/CaH (CBA) and C57BL/6 (C57) mice were exposed to Fe ions and Si ions at the NASA Space Radiation Laboratory. Overall, survival was significantly reduced in CBA compared to C57 following Fe and Si ion irradiation. Cytogenetic analysis using the non-clonal assay for the CBA mouse strain following Fe ion irradiation showed an obvious dose dependent instability at all doses except the lowest dose point. In contrast, chromosome instability was induced at all doses in stem cells derived from C57 mice at day 2. Instability was maintained at the day 6 time point in a dose dependent manner but to a lesser extent with the exception of the highest dose. Cytogenetic analysis using the non-clonal assay for the CBA mouse strain after Si ion irradiation showed chromosome instability observed at all doses at day 2. However, a different trend of instability was observed at the day 6 time point. Similar to the CBA mice, chromosome instability was induced at all doses in stem cells derived from C57 mice at day 2. In contrast to CBA, however, instability at day 6 was maintained in a dose dependent manner but to a lesser extent for all doses when compared to day 2. In conclusion, the study using bone marrow cells obtained from both CBA/CaH (CBA) and C57BL/6 (C57) mouse strains, which differ in their sensitivity to ionizing radiation, suggested that high-LET heavy ions induced two main types of response to irradiation: 1) responses associated with survival of damaged cells and 2) responses associated with the induction of non-clonal chromosomal instability in the surviving progeny of stem cells. These responses were influenced by radiation type and dose. It is also possible that these phenotypic differences were influenced by the genotype of stem cells and/or the haemopoietic microenvironment (i.e. stromal cells). In collaboration with UT Southwestern Medical Center, chromosome aberrations were also investigated in the transformed clones of human lung epithelial cells after exposure to Fe ions. The transformed colonies were collected by researchers at UT Southwestern and cells from the colonies were analyzed for chromosome aberrations using multi-color fluorescent in situ hybridization (mFISH). The karyotype of the parental cells showed specific chromosome aberrations. The transformed clones exhibited an average chromosome number that is different from the parental cells, and some clones showed specific chromosome translocations.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Zhang Y, Uhlemeyer J, Hada M, Asaithamby A, Chen DJ, Wu H. "Proximity within interphase chromosome contributes to the breakpoint distribution in radiation-induced intrachromosomal exchanges." Life Sciences in Space Research. 2014 Jul;2:23-8. http://dx.doi.org/10.1016/j.lssr.2014.05.004 , Jul-2014

Articles in Peer-reviewed Journals	Yuasa-Sunagawa M, Zhang Y, Yeshitla SA, Kadhim M, Wilson BL, Wu H. "mBAND analysis of early and late damages in the chromosome of human lymphocytes after exposures to gamma rays and Fe ions." J Radiat Res. 2014 Mar 1;55(Suppl 1):i87-i88. (Proceedings of Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013.) http://dx.doi.org/10.1093/jrr/rrt182 , Jan-2014